

Reasoning Model Benchmarking — Constraint-Based

Prompting Strategy

Solve this problem using the given constraints: $A < B$, $B < C$, $D < A$, $C < E$. Here, $X < Y$ means X is positioned to the left of Y . Determine the complete ordering.

Benchmark Question

Five students — A , B , C , D , and E — are standing in a line.

A is standing to the left of B .

C is standing to the right of B .

D is standing to the left of A .

E is standing to the right of C .

Expected Order

$D \rightarrow A \rightarrow B \rightarrow C \rightarrow E$

Reasoning

D is left of A ; A is left of B ; B is left of C ; and C is left of E . Therefore: $D < A < B < C < E$.

Observed Output in This Experiment

$D \rightarrow A \rightarrow B \rightarrow C \rightarrow E$

Result

Correct for this benchmark instance.